Spruce Lake Water Treatment Plant

1 Basis of Design

The Spruce Lake Water Treatment Plant is sized for a design flow of 18.2 MGD (average day) and a peak flow of 38.4 MGD on the maximum day, serving a population of 69,135 persons.

Raw water turbidity reaches 24.7 NTU (95th percentile), measured over the last five years. Finished water turbidity shall not exceed 0.12 NTU in 95% of monthly samples.

2 Building Services

The administration building's air handling units are served as follows.

Table 3. Filters

	Filters
Number of filters	36
Area per filter (ft²)	6.9
Filtration rate (cfm/ft²)	346

3 Chemical Systems

Alum is dosed at 17.4 mg/L on the average day and up to 63.3 mg/L on the maximum day. Chlorine is fed at 3.3 mg/L ahead of the clearwell.

4 Process Units

The rapid mix basin provides a detention time of 38 s at design flow, with a velocity gradient of 890 1/s. Flocculation follows in 3 stages, for a total detention time of 20.5 min at design flow.

There are 4 sedimentation basins, each 164 ft long and 46 ft wide, with a side water depth of 12.4 ft; the surface loading rate is 0.53 gpm/ft² at design flow.

Granular filtration uses 10 granular media beds of 599 ft² each. With one bed out of service, the filtration rate is 2.3 gpm/ft².

Table 1. Raw water pumps

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)
P-101A	7,405	83.5	204
P-101B	8,999	97.3	100
P-101C	4,896	77.6	196

5 Disinfection

Two options were compared for primary disinfection.

Table 2. Disinfection options

	Option A: UV	Option B: chlorine contact
Capital cost ($M)	4.15	2.28
Annual energy use (MWh/yr)	559	n/a
Contact time at peak flow (min)	n/a	42